SURAJ DEEPAK JHA

AI Engineer — Generative AI · LLM Systems · Voice AI

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SUMMARY

AI Engineer specializing in Generative AI, LLM systems, and real-time Voice AI, with production experience building scalable AI applications for the insurance domain. Experienced in LLM orchestration, RAG, agentic workflows, FastAPI, and real-time speech pipelines, including production systems handling 20,000+ calls/day.

EXPERIENCE

Probus Insurance Broker Private Limited — **Aug. 2025 – Present**

AI Engineer — *Ahmedabad, Gujarat*

- **Architected and deployed** a production AI voice calling agent handling 20,000+ outbound calls/day for insurance policy renewals, using Pipecat, FastAPI, Twilio, and Sarvam AI with a real-time STT → VAD → LLM → TTS pipeline and sub-second turn latency.
- **Built a provider-agnostic LLM orchestration layer** spanning multiple providers (OpenAI, Anthropic Claude, Google Gemini, Groq, Mistral, DeepSeek, and others) with automatic health-based failover and live mid-call provider switching, powering intelligent analysis, decision support, and agentic workflow automation.
- **Developed vehicle data mapping and normalization pipelines** using Python, Pandas, and RapidFuzz with rule-based scoring and insurance-domain validation logic for GCV and PCV quotation workflows.
- **Designed MCP server architectures and n8n automation pipelines** orchestrating LLMs, REST APIs, and PostgreSQL/Redis databases into scalable production workflows.
- **Built RESTful microservices** with FastAPI and Flask; automated data extraction from 74+ insurance partner portals using Selenium and async HTTP with rate-limit-safe execution.
- **Operated production AI systems** serving thousands of daily user interactions with high availability, using async circuit breakers, Redis-based distributed state, and Prometheus + Grafana observability.

KEY PROJECTS & INITIATIVES

Voice AI Platform — Architecture & Scale

Pipecat | FastAPI | Twilio | Multi-LLM + Multi-STT/TTS Providers

- Implemented a deterministic state machine resolving ~80% of turns without LLM inference (<50ms latency), plus a per-bot RAG pipeline — OpenAI embeddings, chunked ingestion, cosine-similarity retrieval over PostgreSQL — with S3-backed source storage.
- Architected production resilience: async circuit breakers, AMD gate, TRAI-compliant DNC enforcement, and a custom 500-name phonetics module for Indian name pronunciation across regional languages.
- Integrated real-time noise cancellation using Silero VAD and DeepFilterNet over Twilio WebSocket audio streams, improving STT accuracy and turn-detection reliability in noisy call environments.
- Engineered a structured prompt system — call state machine, Hindi objection-handling playbook, anti-hallucination guardrails, and a pronunciation dictionary — to ensure consistent, on-brand agent behavior across languages, refined iteratively from real call recordings.
- Built the enterprise control plane — a multi-tenant Angular + .NET dashboard with role-based access control, audit trails, live monitoring, and a function-calling tool framework — deployed via Docker with multi-worker scaling.

Intelligent Vehicle Data Mapping System — GCV + PCV

Python | RapidFuzz | PostgreSQL | Rule-Based Pipeline

- Built an end-to-end vehicle master data standardization pipeline processing thousands of records across 15+ insurance partners (HDFC, TATA, ICICI, Bajaj, Digit, SBI, and others) using fuzzy matching, multi-field rule gates (vehicle type, fuel, GVW), and cross-field confidence scoring.
- Directly powers the company's quotation portal — the foundational system enabling premium generation for GCV and PCV insurance lines — with modular per-insurer engines and a PostgreSQL audit trail.

Real-Time API Health Intelligence Platform

FastAPI | asyncio | aiohttp | Streamlit | Prometheus

- Built a monitoring platform concurrently tracking 74+ insurance partner APIs using Python asyncio + aiohttp, with status classification (UP/DOWN/SLOW/ERROR), multi-channel alerting (email, Slack, Teams), and smart state-change suppression.
- Delivered a real-time FastAPI dashboard (SSE) and a Streamlit dashboard (Plotly charts) with per-API isolation, configurable timeouts, and exponential backoff retries to minimize cascading failures.

LLM Applications & MCP Workflow Automation

OpenAI GPT-4o | Anthropic Claude | Gemini | Ollama | n8n | MCP

- Built and deployed multiple LLM-powered business applications using OpenAI GPT-4o, Anthropic Claude, Google Gemini, and Ollama (Llama 3) for document analysis, decision support, and automated customer interaction workflows.
- Designed MCP server architectures exposing tools and resources to LLMs for structured reasoning over insurance data; integrated with n8n to orchestrate multi-step agentic pipelines across APIs, databases, and third-party services.
- Applied prompt engineering techniques (few-shot prompting, structured outputs, tool calling, prompt evaluation) to improve LLM accuracy on insurance-domain tasks; benchmarked outputs across GPT-4o, Claude, and Gemini for production suitability.

TECHNICAL SKILLS

Languages: Python, SQL, C++, C#, TypeScript

AI / ML: Generative AI, LLMs, RAG, NLP, Machine Learning, Deep Learning, Prompt Engineering, Fine-Tuning (LoRA, QLoRA), Agentic AI, Real-Time Voice AI

Voice AI Framework: Pipecat, Twilio Media Streams, Silero/WebRTC VAD

STT/TTS Providers: Sarvam AI, Deepgram, ElevenLabs, Cartesia, Azure Speech, Google Speech

LLM Providers: OpenAI, Anthropic Claude, Google Gemini, Groq, Mistral, DeepSeek, OpenRouter, Sarvam AI

Vector Search / RAG: OpenAI Embeddings (text-embedding-3-small), Cosine-Similarity Retrieval, Semantic Chunking, PostgreSQL-Backed Knowledge Bases

Backend & APIs: FastAPI, Flask, REST APIs, WebSockets, AsyncIO, Uvicorn, Gunicorn

Engineering: Microservices, Distributed Systems, System Design, Async Programming

Agentic & Automation: MCP (Model Context Protocol), Function Calling / Structured Outputs, n8n, Selenium

Enterprise / Full-Stack: Angular, .NET, Entity Framework Core, RBAC & Multi-Tenant Architecture

Libraries: NumPy, Pandas, Scikit-learn, TensorFlow, RapidFuzz, OpenCV

Infra & DevOps: Docker, Docker Compose, Git, Linux, CI/CD, Prometheus, Grafana, Structured Logging (structlog)

Cloud: AWS (S3), Azure

Databases: PostgreSQL, MySQL, SQLite, Redis

Tools: GitHub, Postman, VS Code, Google Colab, PyCharm

EDUCATION

University of Mumbai — Master of Computer Applications **Sept. 2024 – May 2026**

Mumbai, India

Nagindas Khandwala College — B.Sc. Computer Science **Jun. 2021 – May 2024**

Mumbai, India

CERTIFICATIONS

- **Cloud Computing and Distributed Systems** — NPTEL (Elite), IIT Kanpur — Score: 76% — Jan–Mar 2026
- **Claude 101 — Anthropic Claude AI Fundamentals** — Anthropic